Generation of Individual Parameters for the Logistic Growth Model

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1 Notation and common observation model

Matrices are denoted by uppercase letters, vectors by lowercase letters, and scalars by lowercase letters, except for the fixed dimension J. Let $n \in \mathbb{N}_{>0}$ be the number of individuals and $J \in \mathbb{N}_{>0}$ the number of observations per individual. The individual parameters are

$$\beta_i = (\beta_{i1}, \beta_{i2}, \beta_{i3})^\top \in \mathbb{R}^3, \qquad i \in \{1, \dots, n\},$$

where $\beta_{i1}, \beta_{i2}, \beta_{i3} \in \mathbb{R}$ denote the asymptote, inflection point, and scale. The reference population parameters are

$$\mu = (\mu_1, \mu_2, \mu_3)^\top \in \mathbb{R}^3, \qquad \Sigma \in \mathbb{R}^{3\times 3}, \quad \Sigma \succ 0, \qquad \sigma^2 \in \mathbb{R}_{>0}.$$

Here $\mu_d \in \mathbb{R}$ and $\Sigma_{de} \in \mathbb{R}$ are scalar entries. All vectors are column vectors. We write $I_d \in \mathbb{R}^{d\times d}$ for the identity matrix and $0_d \in \mathbb{R}^d$ for the zero vector.

For observation times $t = (t_1, \dots, t_J)^\top \in \mathbb{R}^J$, generate

$$y_{ij} = f(t_j, \beta_i) + e_{ij}, \qquad f(t_j, \beta_i) = \frac{\beta_{i1}}{1 + \exp\Big(-\frac{t_j - \beta_{i2}}{\beta_{i3}}\Big)}, \qquad e_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \tag{1}$$

where $t_j, y_{ij}, e_{ij}, f(t_j, \beta_i) \in \mathbb{R}$. The scalar errors are independent of the latent-generation draws. For each individual,

$$y_i = (y_{i1}, \dots, y_{iJ})^\top \in \mathbb{R}^J, \qquad e_i = (e_{i1}, \dots, e_{iJ})^\top \in \mathbb{R}^J.$$

The complete latent and observation matrices are

$$B = \begin{pmatrix} \beta_1^\top \\ \vdots \\ \beta_n^\top \end{pmatrix} \in \mathbb{R}^{n\times 3}, \qquad Y = \begin{pmatrix} y_1^\top \\ \vdots \\ y_n^\top \end{pmatrix} \in \mathbb{R}^{n\times J}, \qquad E = \begin{pmatrix} e_1^\top \\ \vdots \\ e_n^\top \end{pmatrix} \in \mathbb{R}^{n\times J}.$$

The function returns the arrays corresponding to (B, Y, t).

The three constructions below replace the Gaussian population law $\beta_i = \mu + b_i$, $b_i \sim \mathcal{N}(0_3, \Sigma)$, where $b_i \in \mathbb{R}^3$, while retaining the observation model. Therefore, μ and Σ need not equal the actual mean and covariance of the generated individual parameters. They remain reference parameters of the respective generators.

2 Coordinate-wise three-component Gaussian mixtures

2.1 Sampling construction

Generate a matrix of candidate centers $M \in \mathbb{R}^{3\times 3}$, with scalar entries

$$M_{kd} = \mu_d + s\eta_{kd}, \qquad \eta_{kd} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1), \qquad s = 50 \in \mathbb{R}_{>0}, \qquad k, d \in \{1, 2, 3\}. \tag{2}$$

Here $\eta_{kd} \in \mathbb{R}$; equivalently the noise matrix $H = (\eta_{kd}) \in \mathbb{R}^{3\times 3}$. The matrix M is sampled once per dataset and shared across individuals.

Independently for each individual and coordinate, draw a scalar component index

$$c_{id} \sim \mathrm{Unif}\{1,2,3\}, \qquad c_{id} \in \{1,2,3\},$$

and assemble the selected center vector

$$a_i = (M_{c_{i1},1}, M_{c_{i2},2}, M_{c_{i3},3})^\top \in \mathbb{R}^3.$$

Let $L \in \mathbb{R}^{3\times 3}$ be the lower-triangular Cholesky factor of Σ, so $LL^\top = \Sigma$. Independently draw

$$\xi_i \sim \mathcal{N}(0_3, I_3), \qquad \xi_i \in \mathbb{R}^3, \qquad \beta_i = a_i + L\xi_i \in \mathbb{R}^3. \tag{3}$$

3 Single-moon construction

This construction combines a bimodal first parameter with a single noisy semicircle in the (β_{i2}, β_{i3}) plane.

3.1 First parameter

Generate the scalar asymptote directly from the mixture

$$\beta_{i1} \sim \tfrac{1}{2}\mathcal{N}(\mu_1 - \delta, \omega^2) + \tfrac{1}{2}\mathcal{N}(\mu_1 + \delta, \omega^2), \qquad \delta = 40 \in \mathbb{R}_{>0}, \quad \omega = 10 \in \mathbb{R}_{>0}. \tag{4}$$

The two component centers are separated by 80 and each component has standard deviation 10. This scalar draw is independent of the moon pair below.

3.2 Second and third parameters

Generate a scalar angle and a two-dimensional noise vector independently:

$$\alpha_i \sim \mathrm{Unif}(0, \pi), \quad \alpha_i \in \mathbb{R}, \qquad \nu_i = (\nu_{i1}, \nu_{i2})^\top \sim \mathcal{N}(0_2, \omega_m^2 I_2), \quad \nu_i \in \mathbb{R}^2,$$

where $\omega_m = 8 \in \mathbb{R}_{>0}$ and $\omega_m^2 I_2 \in \mathbb{R}^{2\times 2}$. With radius $\rho = 60 \in \mathbb{R}_{>0}$, set

$$\begin{pmatrix} \beta_{i2} \\ \beta_{i3} \end{pmatrix} = \begin{pmatrix} \mu_2 \\ \mu_3 \end{pmatrix} + \rho \begin{pmatrix} 1 - \cos\alpha_i \\ \frac{1}{2} - \sin\alpha_i \end{pmatrix} + \nu_i \in \mathbb{R}^2. \tag{5}$$

Equivalently, the scalar coordinate equations are

$$\beta_{i2} = \mu_2 + 60(1 - \cos\alpha_i) + \nu_{i1}, \qquad \beta_{i3} = \mu_3 + 60(\tfrac{1}{2} - \sin\alpha_i) + \nu_{i2}.$$

Finally assemble $\beta_i = (\beta_{i1}, \beta_{i2}, \beta_{i3})^\top \in \mathbb{R}^3$.

Before noise is added, the last two coordinates describe the lower semicircle of radius 60 centered at $(\mu_2 + 60, \mu_3 + 30)^\top \in \mathbb{R}^2$. Uniform angle gives uniform arc-length sampling, while the Gaussian noise thickens this arc into a smooth moon-shaped density.

4 Empty-ball construction: a uniform spherical shell

4.1 Sampling construction

Let the inner and outer radii be scalar constants

$$r_{\mathrm{in}} = 30 \in \mathbb{R}_{>0}, \qquad r_{\mathrm{out}} = 80 \in \mathbb{R}_{>0}.$$

Independently draw a Gaussian vector and a uniform scalar:

$$g_i \sim \mathcal{N}(0_3, I_3), \quad g_i \in \mathbb{R}^3, \qquad u_i \sim \mathrm{Unif}(0,1), \quad u_i \in \mathbb{R}.$$

Construct the direction vector and scalar radius

$$d_i = \frac{g_i}{\|g_i\|_2} \in \mathbb{R}^3, \quad \|d_i\|_2 = 1, \qquad s_i = \left[r_{\mathrm{in}}^3 + u_i(r_{\mathrm{out}}^3 - r_{\mathrm{in}}^3)\right]^{1/3} \in [r_{\mathrm{in}}, r_{\mathrm{out}}]. \tag{6}$$

The generated individual parameters are

$$\beta_i = \mu + s_i d_i \in \mathbb{R}^3. \tag{7}$$

The direction is uniform on the unit sphere; the cube-root transformation samples uniformly in shell volume, not uniformly in radial distance.